Sokoine University of Agriculture (SUA)
Department of Agricultural Engineering
Electronics and Precision Agriculture Lab (EPAL)

AGE 418
Sensors and Controls in Precision Agriculture

Lecture 12 Compendium:
Precision Irrigation and Water Management
Optimizing the Most Critical Resource

Instructor: Dr. Kadeghe Fue
Academic Year: 2025

Contents

1 1. The Global Water Challenge **2**

2 2. Defining Precision Irrigation **2**

3 3. Irrigation Systems and Control **2**
3.1 1. Drip Irrigation (The Gold Standard) 2
3.2 2. Variable Rate Irrigation (VRI) for Center Pivots 2

4 4. Variable Rate Mechanics: Pulse Width Modulation (PWM) **3**
4.1 The Duty Cycle 3

5 5. Scheduling Strategies: The "Brains" **3**
5.1 Strategy A: Soil-Based (The "Fuel Gauge") 3
5.2 Strategy B: Weather-Based (The "Bank Account") 3
5.3 Strategy C: Plant-Based (Asking the Plant) 4

6 6. Automation and IoT **4**
6.1 Components 4
6.2 Benefits of Automation 4

7 7. Lecture Assessment **5**
7.1 Part A: Standard Concept Questions 5
7.2 Part B: Advanced Analysis (Open Source Data) 5

1 1. The Global Water Challenge

Agriculture consumes approximately **70% of global freshwater withdrawals**. As competition for water increases from urbanization and industrialization, agriculture must produce "More Crop Per Drop."

In Tanzania, major river basins like the **Rufiji** and **Pangani** are under stress. Traditional irrigation methods (Flood/Furrow) often have efficiencies as low as 40-60%. Precision Irrigation aims to increase this to **90-95%**.

2 2. Defining Precision Irrigation

Precision Irrigation is the application of water at the **Right Rate**, **Right Time**, and **Right Place** to meet the specific needs of individual plants or management zones, rather than the whole field uniformly.

The Core Question: "When and How Much?"

Traditional irrigation follows a rigid schedule (e.g., "Every Tuesday"). Precision irrigation follows **Demand**:

- **When?** When the soil moisture drops below the "Management Allowable Depletion" (MAD) point.
- **How Much?** Enough to refill the root zone to Field Capacity, but not past it (to avoid leaching).

3 3. Irrigation Systems and Control

To implement precision irrigation, we need controllable hardware.

3.1 1. Drip Irrigation (The Gold Standard)

Delivers water directly to the root zone via emitters.

- **Precision Potential:** Very High. Zones can be turned on/off individually using solenoid valves.
- **Efficiency:** >95% (Zero wind drift, near-zero evaporation).
- **Automation:** Easily controlled by simple timers or IoT nodes.

3.2 2. Variable Rate Irrigation (VRI) for Center Pivots

Center pivots irrigate large circles (up to 100 ha). Traditionally, they apply a uniform rate (e.g., 10mm) across the whole circle.

- **Zone Control VRI:** The pivot speed is adjusted. It moves fast over wet areas (applying less water) and slow over dry areas (applying more).
- **Nozzle Control VRI:** Individual sprinklers are pulsed ON/OFF using solenoid valves to vary the rate along the boom span.

4 4. Variable Rate Mechanics: Pulse Width Modulation (PWM)

How do we vary the flow rate of a nozzle without changing the pressure (which would ruin the droplet pattern)? We use **PWM**.

4.1 The Duty Cycle

A solenoid valve opens and closes rapidly (e.g., on a 60-second cycle).

- **100% Duty Cycle:** Valve open 60s (Full Rate).
- **50% Duty Cycle:** Valve open 30s, closed 30s (Half Rate).
- **0% Duty Cycle:** Valve closed (Zero Rate - e.g., crossing a road/creek).

5 5. Scheduling Strategies: The "Brains"

Having a VRI machine is useless if you don't know *when* to turn it on. There are three main scheduling strategies.

5.1 Strategy A: Soil-Based (The "Fuel Gauge")

Measuring the water actually in the soil "tank."

- **Sensors:** Capacitance probes or Tensiometers (Lecture 4).
- **Logic:** IF $Moisture < Threshold$ THEN $Pump = ON$.
- **Pros:** Direct measurement.
- **Cons:** A sensor only measures a tiny volume of soil. Is it representative of the whole 50 hectares?

5.2 Strategy B: Weather-Based (The "Bank Account")

Balancing inputs and outputs.

- **Inputs:** Rain + Irrigation.
- **Outputs:** Evapotranspiration (ET_c).
- **Formula:** $ET_c = ET_o \times K_c$
  - ET_o: Reference Evapotranspiration (from a Weather Station).
  - K_c: Crop Coefficient (varies by growth stage).
- **Pros:** Good for regional management. Low hardware cost.
- **Cons:** Mathematical estimation, not direct measurement.

5.3 Strategy C: Plant-Based (Asking the Plant)

Measuring plant stress directly.

- **Sensors:** Infrared Thermometers (Canopy Temperature), Dendrometers (Stem shrinkage).
- **Principle:** A well-watered plant transpires and cools itself. A stressed plant closes stomata and heats up.
- **Metric:** Crop Water Stress Index (CWSI).

6 6. Automation and IoT

Modern irrigation uses **Closed-Loop Control**.

6.1 Components

1. **IoT Node:** Solar-powered box in the field reading soil moisture. 2. **Gateway:** LoRaWAN base station receiving data. 3. **Cloud Server:** Runs the logic ("Is it too dry? Is rain forecast?"). 4. **Controller:** Relays at the pump house to switch the pump.

6.2 Benefits of Automation

- **Labor Saving:** No need to drive to the field to open valves.
- **Night Irrigation:** Can irrigate at 2 AM to reduce evaporation losses.
- **Optimization:** Prevents over-watering (leaching nutrients) and under-watering (yield loss).

7 7. Lecture Assessment

7.1 Part A: Standard Concept Questions

1. Define **Precision Irrigation**. How does it differ from traditional "Calendar Scheduling"?
2. Explain the concept of **Variable Rate Irrigation (VRI)** for a Center Pivot.
3. Describe the **Pulse Width Modulation (PWM)** technique for controlling nozzle flow rates. Why is it better than simply changing pressure?
4. List the three primary methods of Irrigation Scheduling. Which one is considered the "Fuel Gauge" approach?
5. Write the formula for calculating Crop Evapotranspiration (ET_c). Define the variables involved.
6. How does a **Thermal Sensor** detect water stress in a crop canopy?
7. What is the **Crop Water Stress Index (CWSI)**? What does a value of 0.9 indicate?
8. Explain the role of **IoT** in irrigation automation. What is the advantage of LoRaWAN for this application?
9. Why is **Subsurface Drip Irrigation (SDI)** considered more efficient than surface drip?
10. Identify one major barrier to adopting VRI systems for smallholder farmers in Tanzania.

7.2 Part B: Advanced Analysis (Open Source Data)

These questions require internet access and the use of open-source datasets/tools.

1. **WaPOR Evapotranspiration Analysis:** Visit the **FAO WaPOR Portal** (https://wapor.apps.fao.org/). Select the **Kamsonga (Mbarali)** area. *Task:* View the "Actual Evapotranspiration and Interception (AETI)" layer for the last growing season. Identify areas of low AETI within the irrigated block. Does this indicate water stress or poor crop stand? Correlate this with the "Biomass" layer.
2. **Designing a Scheduling Algorithm:** You are programming a Smart Controller using Python logic. *Task:* Write pseudocode for the controller. *Inputs:* Soil Moisture (*SM*, %), Rainfall Forecast 24h (*RF*, mm), Crop Stage (*Stage*, "Veg" or "Mature"). *Rules:*
   - Maintain SM between 30% and 40%.
   - If RF > 5mm, do NOT irrigate (Save water).
   - If Stage is "Mature" (drying down), stop irrigation.
3. **Cost-Benefit of VRI:** A 50-hectare pivot in Iringa costs TZS 200 Million. Adding VRI hardware costs an extra TZS 30 Million. *Task:* The field has sandy ridges (20% of area) and clay bottoms (80%). Without VRI, you over-water the clay to satisfy the sand. With VRI, you save 15% of total water and energy. If water pumping costs TZS 100,000 per hectare per season, calculate the payback period for the VRI investment.

4. **OpenWeatherMap API for ET:** Visit **OpenWeatherMap.org**. Look at the "One Call API 3.0". *Task:* Identify which parameters from the API (Temp, Humidity, Wind Speed, Solar Radiation) are needed to calculate **Reference Evapotranspiration (ET_o)** using the Penman-Monteith equation. Is any parameter missing?

5. **Satellite Soil Moisture (SMAP/Sentinel-1):** Search for **NASA SMAP** or **Sentinel-1 Soil Moisture** products on Google Earth Engine. *Task:* The spatial resolution of SMAP is 9km. The resolution of Sentinel-1 is 10m. Which product is suitable for a district-level drought warning system, and which is suitable for farm-level irrigation management? Explain why.